

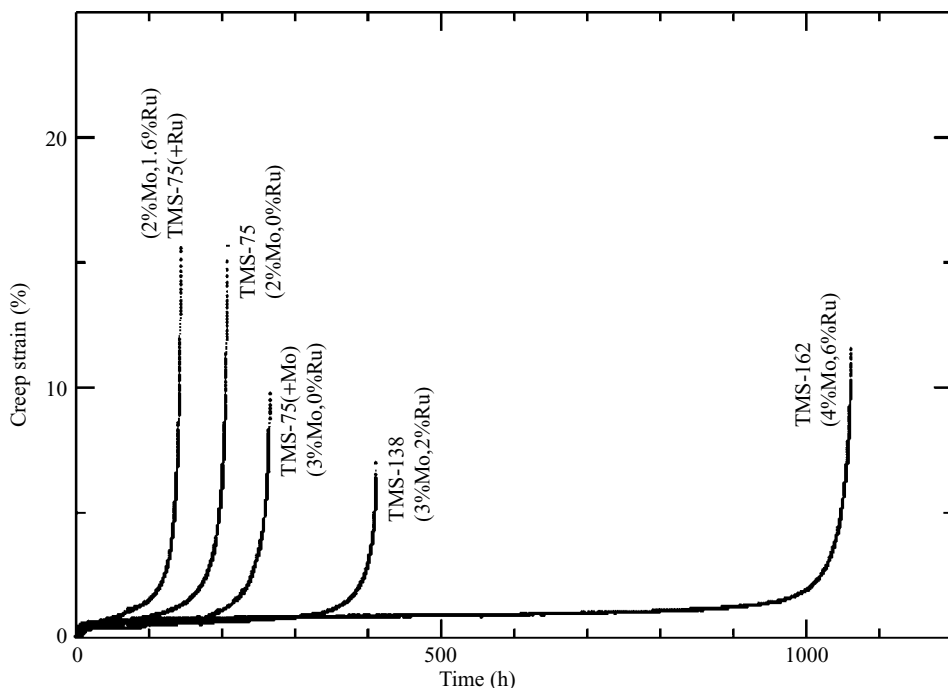
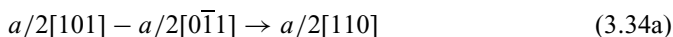
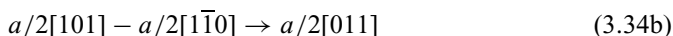


Fig. 3.64. Creep curves of the superalloys TMS-75, TMS-75(+Ru), TMS-75(+Mo), TMS-138 and TMS-162 tested at 1100 °C and 137 MPa [64,65].

normal, slip-deposited dislocations form two perpendicular sets, for example of $a/2[0\bar{1}1]$ dislocations, one resulting from slip on (111) and the other on $(\bar{1}\bar{1}1)$. These can react to form a single set with average line vector $[100]$. Further $a/2(1\bar{1}0)$ dislocations resulting from slip on the other two octahedral planes, $(1\bar{1}1)$ and $(\bar{1}\bar{1}1)$, can then be knitted into the net, consistent with reactions of the type [63]



and



or, alternatively,



Thus it is clear that the equilibrium network can arise only when sufficient creep dislocations have formed on all four of the $\{111\}$ -type planes. Harada and co-workers have isolated a strong correlation between the dislocation density of the equilibrium network in single-crystal superalloys of varying compositions and their creep performance under loading at 1100 °C and 137 MPa (see Figures 3.64 and 3.65), with the dislocation spacing being found to be proportional to the logarithm of the minimum creep rate [64,65]. Use of this effect has been made in designing TMS-162, which is one of the strongest single-crystal superalloys

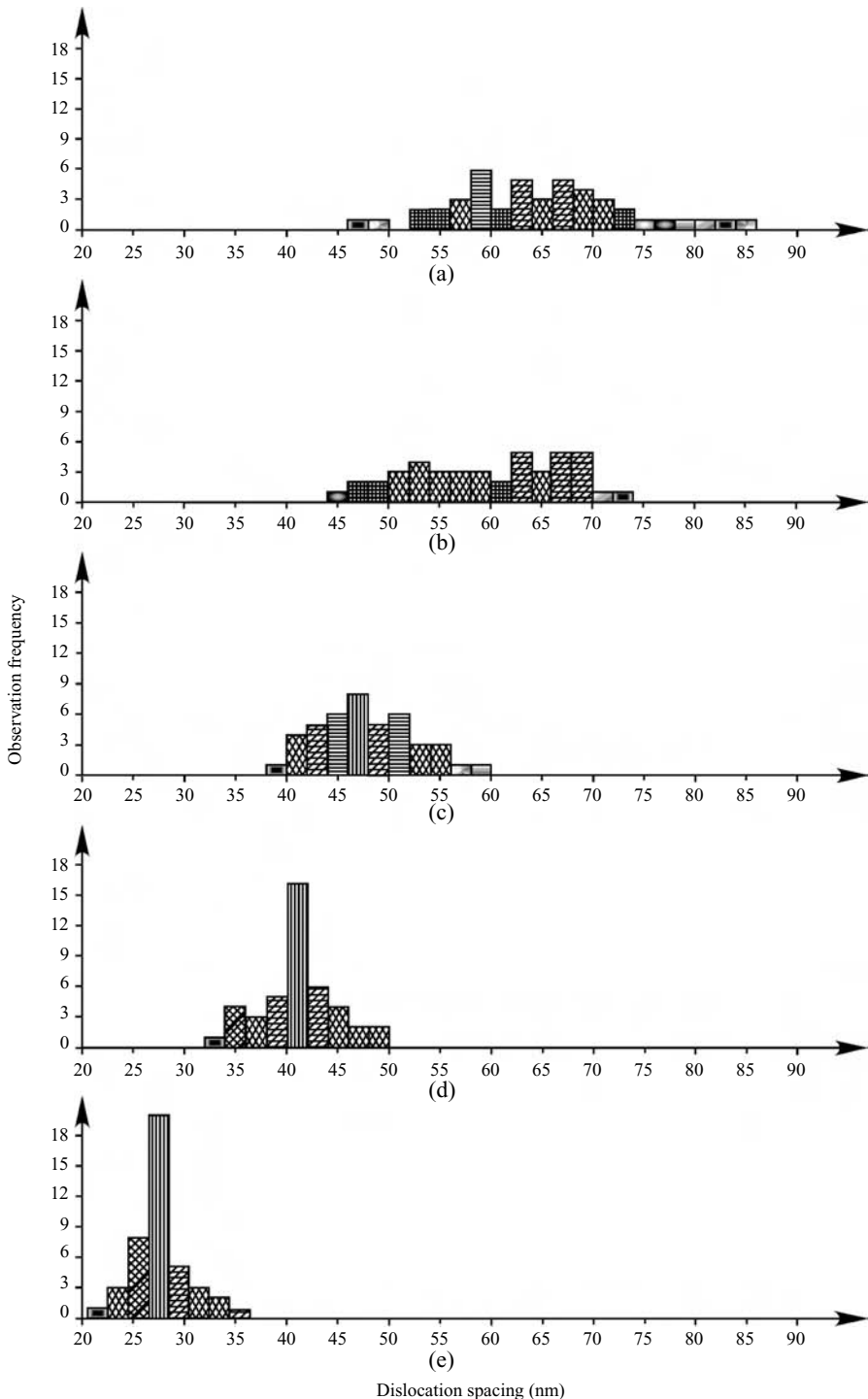


Fig. 3.65. Measurements of the equilibrium γ/γ' dislocation spacings in (a) TMS-75(+Ru), (b) TMS-75, (c) TMS-75(+Mo), (d) TMS-138 and (e) TMS-162 tested at 1100 °C and 137 MPa [64,65]; taken with the data in Figure 3.64, a strong creep strengthening effect is confirmed when the dislocation spacing is small.

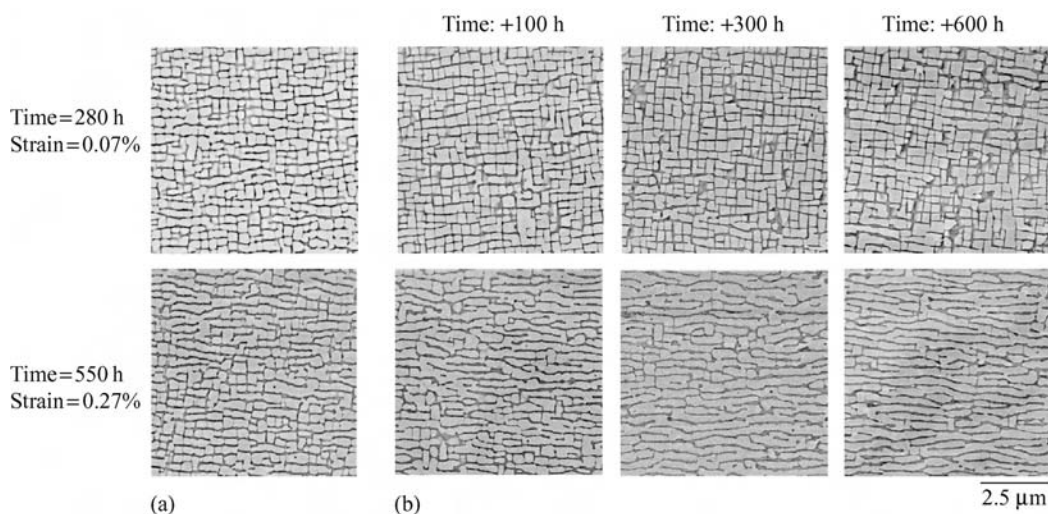


Fig. 3.66. SEM micrographs [62] of the CMSX-4 superalloy deformed at 950 °C. (a) γ/γ' microstructures after 280 h (0.07% strain) and 550 h (0.27% strain), with an applied stress of 185 MPa. (b) Microstructures after further heat treatment of 100, 300 and 600 h, with zero applied stress.

to date (see Figure 1.17). Additions of Mo are critical to the superior creep performance displayed, since the lattice misfit is reduced as the concentration of this element increases.

The relaxation of interfacial misfit by the above mechanism is important since it is now recognised that rafting occurs only once this has occurred [60–62]. Consider a negatively misfitting alloy such as CMSX-4. The micromechanism of creep deformation in the tertiary creep regime involves the multiplication of dislocations in γ , initially in the horizontal channels for reasons discussed earlier; the relaxation of interfacial misfit thus occurs preferentially on the γ/γ' interfaces whose normals lie parallel to the direction of loading. Rafting requires diffusion of γ -partitioning elements such as Cr, Re out of the vertical channels and γ' -partitioning elements such as Al, Ta into them; the evidence indicates that such mass transport can occur only once the lattice misfit has become relaxed, probably because diffusion occurs near or within the relaxed interfaces. The possibility of pipe diffusion of solutes within the dislocation networks should not be dismissed, although no direct evidence for this has been found as yet. The role of interfacial misfit relaxation in the rafting phenomenon has been confirmed with experiments which have demonstrated that partially crept single crystals continue to raft *even after the externally applied load is removed* [60,62]. Moreover, the kinetics of rafting are not slowed in any way by the removal of the stress field. Figure 3.66 shows the results of experiments on CMSX-4 [62]. For deformation at 950 °C and 185 MPa, an accumulated strain of about 0.1% is sufficient to relax the interfacial misfit and, thereafter, rafting without the externally applied load continues at the same rate as with the load present.

A controversial question concerns the influence of the rafting effect on the creep capability of the single-crystal superalloys. Note (see Figure 3.61) that as many as ten or more γ' precipitates coalesce in the transverse direction; thus a fully developed raft is formed from 100 or more precipitates of total cross-sectional area equal to several hundreds of square

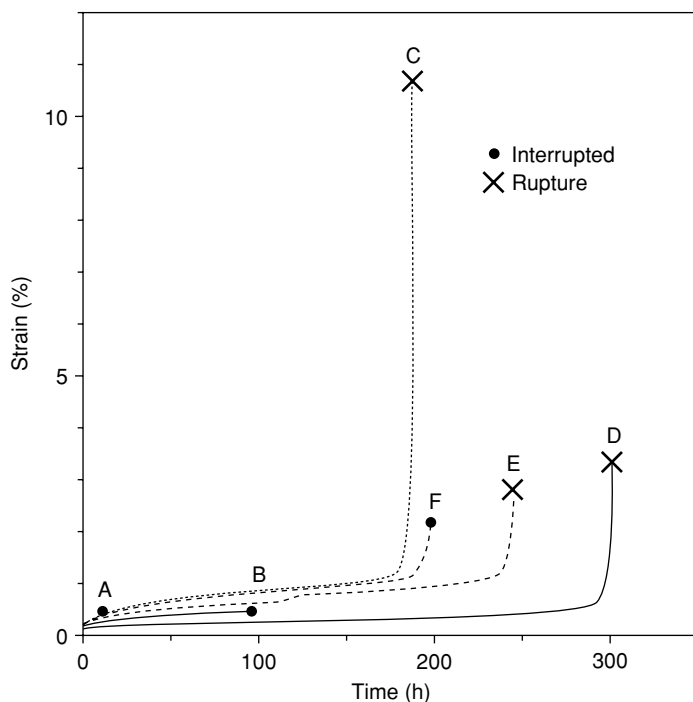


Fig. 3.67. Constant load creep data for the CMSX-4 superalloy, deformed at 1150 °C and 100 MPa.

micrometres, since the rafts have a thickness equal to one γ' particle. Extensive rafting leads eventually to the inversion of the γ/γ' structure such that the continuous, matrix phase is γ' rather than γ . Under these circumstances, rafting reduces the activity of the $a/2\langle 1\bar{1}0\rangle\{111\}$ dislocations in γ . This is supported by various pieces of evidence [66]. First, extrapolation of the tertiary creep law to temperatures where rafting occurs leads to an underestimate of the creep performance, particularly at small applied stresses when creep life is long; see Figure 3.54(d). Second, in the rafting regime, fully developed rafts are found to have formed at times equivalent to the onset of, or even before, the plateau observed in the creep curves at these very high temperatures; see Figure 3.51(c). Thus rafting is at least partially responsible for the creep-hardening effect which occurs initially and the creep plateau that occurs thereafter. Third, there is considerable scatter in the creep rupture lives at these very high temperatures (see Figure 3.67); however, the creep plateau is very reproducible. Final rupture at these temperatures is associated with extensive creep cavitation at porosity introduced either during the casting process into the interdendritic regions [67], or alternatively at pores forming around TCP phases which precipitate at these elevated temperatures [66]. Cavitation is, however, restricted to those pores present in the vicinity of fracture surface (see Figure 3.68), and it is probable therefore that their formation is strain-controlled. Hot isostatic pressing has, however, been found to increase the mean value of the rupture life somewhat. However, more research is needed to clarify the reasons for the onset of the rapid acceleration in creep strain rate leading to rupture, and the micromechanical deformation modes which lead to the cutting of the rafted γ' microstructure.

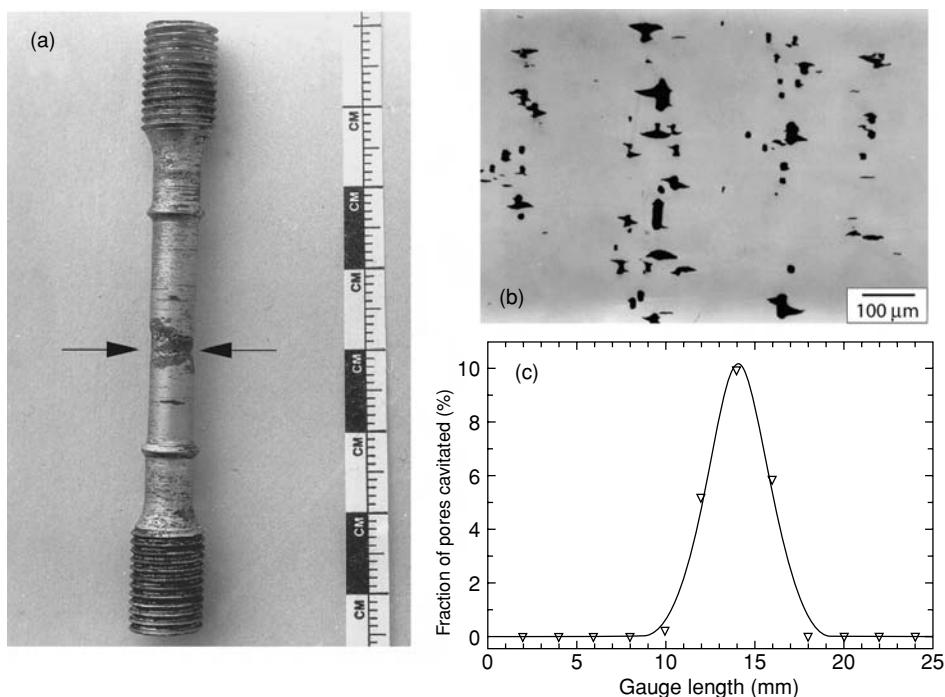


Fig. 3.68. CMSX-4 creep testpiece interrupted after 2.2% strain after 200 h at 1150 °C and 100 MPa (see the creep curve marked 'F' in Figure 3.67). (a) Necked creep specimen at point of fracture. (b) Micrograph of creep-cavitated casting porosity in necked region. (c) Variation of fraction of pores cavitated as a function of distance along specimen, providing confirmation that cavitation is restricted to the necked region.

3.3.2 Performance in fatigue

As for all metallic materials, the single-crystal superalloys are prone to fatigue failure due to oscillating loads, at stresses which are nominally elastic. This is because plastic deformation can occur in a very localised sense, typically near sites of stress concentration; this fatigue damage is most often in the form of intense, localised slip with the dislocation activity restricted to a small number of lattice planes [68]. This phenomenon is often referred to as 'cyclic slip localisation'; particularly at temperatures of around 700 °C and when stresses are high this encourages the formation of 'persistent slip bands' (PSBs) which shear the γ' particles [69]; see Figure 3.69. Fatigue initiation can always be traced to sites of stress concentration, for example casting pores or machining marks. At higher temperatures and when stresses are low, dislocation activity is restricted to the γ channels and cyclic slip localisation is less intense; thus deformation is more homogeneous, but nonetheless limited to the most highly stressed $\{111\}$ planes [70]; see Figure 3.70. Under these conditions, fatigue initiation occurs at surface pits or cracks in the oxide films – and the life is improved markedly if protective coatings (for example the aluminides) are used.

In practice, the environment experienced by the blading in service is very difficult to replicate exactly in the laboratory. Consequently, reliance is placed on the characterisation of

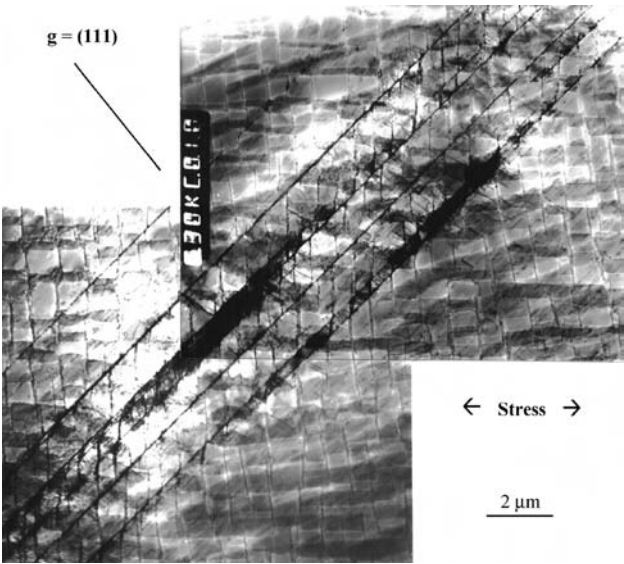


Fig. 3.69. Persistent slip bands in a $\langle 001 \rangle$ -oriented single-crystal superalloy cycled at 700°C [69], in which the γ' particles are sheared.

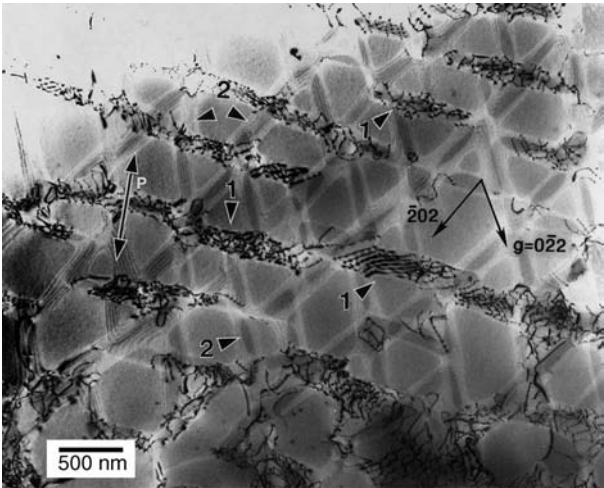


Fig. 3.70. Dislocation microstructure in the AM1 single-crystal superalloy deformed at 950°C with $\Delta\epsilon_{\text{total}} = 1.27\%$ and with $R = 0$ [70]. P is the projection of the loading axis. Dislocations are localised at $\{001\}$ interfaces, see arrows marked 1; $\{010\}$ and $\{100\}$ channels, see arrows marked 2, are free of dislocations.

fatigue performance under conditions which can be easily controlled, for example isothermal tests under uniaxial load or strain control. The cyclic stress range, $\Delta\sigma$, is defined as $\sigma_{\text{max}} - \sigma_{\text{min}}$, where σ_{max} and σ_{min} are the maximum and minimum stresses seen during the cycle, and the mean stress, σ_{mean} , is $\frac{1}{2}(\sigma_{\text{max}} + \sigma_{\text{min}})$. The R -ratio is defined as $\sigma_{\text{min}}/\sigma_{\text{max}}$, so that $R = -1$ corresponds to fully reversed loading with $\sigma_{\text{max}} = -\sigma_{\text{min}}$ and $R = 0$ corresponds

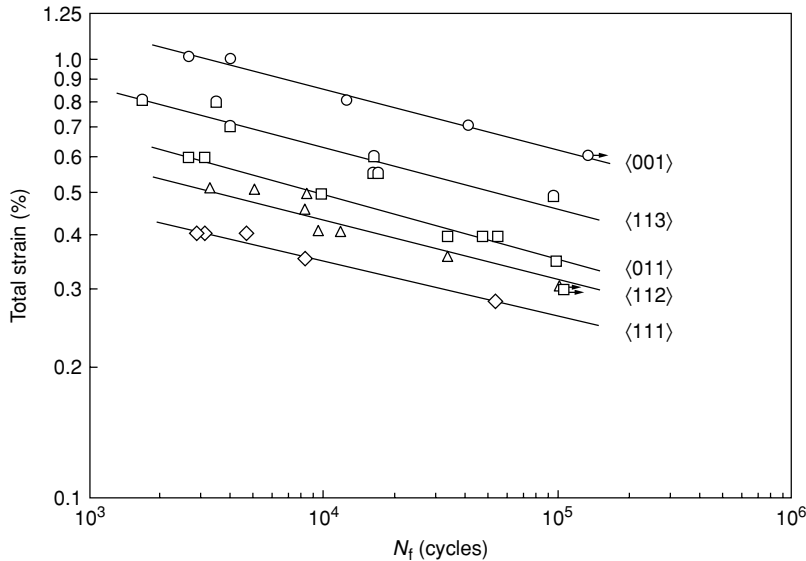


Fig. 3.71. The orientation dependence of the strain-controlled LCF life of an experimental single-crystal superalloy at 980 °C, for various values of the total strain range, $\Delta\epsilon_{\text{total}} = 0.6$ [71]. Axial load: $K_T = 1.0$; frequency = 0.33 Hz.

to pure cyclic conditions with $\sigma_{\min} = 0$. Two distinct forms of fatigue failure are identified. Low-cycle fatigue (LCF) arises when the stress amplitude places the nominal stresses at or beyond the elastic limit – failure occurs typically within 10^5 cycles, with a majority of the fatigue life being spent in the propagation (rather than initiation) stage. From a practical perspective, the driving force for LCF damage originates from abrupt changes in loading due to engine start-up and stopping. Conversely, high-cycle fatigue (HCF) arises from smaller stress amplitudes, typical of those imposed on the blading by vibrations or resonance. In this case, the nominal stresses are always in the elastic regime, and the majority of the fatigue life is spent in the initiation stage; failure occurs in excess of 10^5 cycles.

A Low-cycle fatigue

Under LCF testing conditions, which are often carried out under total strain control, it is found that the fatigue performance depends very strongly on the orientation of the loading axis with respect to the crystallographic orientation, i.e. considerable anisotropy is displayed. Figure 3.71 shows data due to Dalal, Thomas and Dardi [71], who measured the LCF response of smooth, polished specimens of an experimental alloy (composition in wt%: Ni–6.8Al–13.8Mo–6W) at 980 °C, under total strain control and with $R = 0$. The logarithm of the total strain range, $\Delta\epsilon_{\text{total}}$, is found to be inversely proportional to the logarithm of the fatigue life, N_f , for any given orientation. Of the three orientations at the corners of the standard stereographic triangle, i.e. <001>, <011> and <111>, the <001> orientation displayed the best performance and <111> the worst; all other orientations were between these two

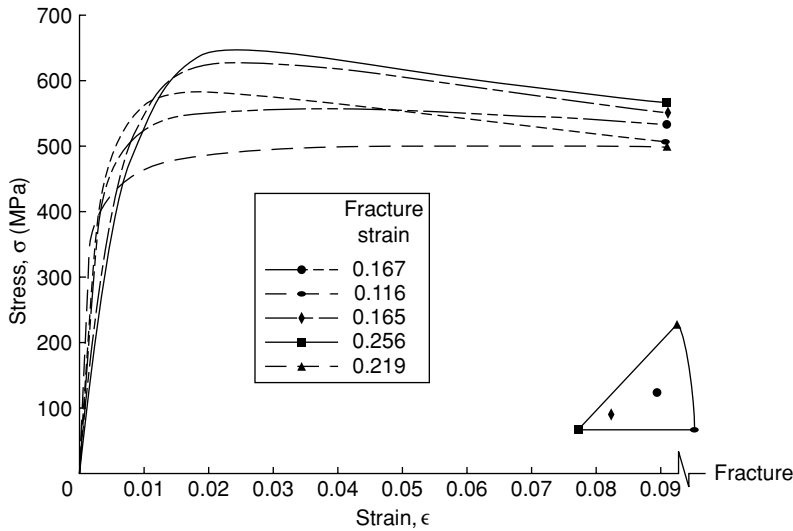


Fig. 3.72. Stress–strain curves for PWA1480 single-crystal testpieces, for various orientations taken from the standard stereographic triangle [72].

limits. Thus the fatigue strength correlates strongly with the elastic modulus in the loading direction – so that a direction such as $\langle 001 \rangle$, which is elastically soft, displays very much better properties than one which is stiff, such as $\langle 111 \rangle$. In fact, it can be shown that if one plots the product $\Delta\epsilon_{\text{total}} E_{\text{hkl}}$ rather than $\Delta\epsilon_{\text{total}}$ against N_f , then the fatigue data collapse onto a straight line. The term E_{hkl} is the plane-specific elastic modulus along the loading axis.

These observations can be rationalised in the following way. Assuming for one moment that the yield stresses in the $\langle 001 \rangle$ and $\langle 111 \rangle$ directions are equal, the elastic strain at yield is smaller for a specimen tested along $\langle 111 \rangle$ on account of its greater stiffness; thus, under total strain control, the plastic strain available to drive the fatigue process is correspondingly greater, so that poorer properties are displayed. In fact, the yield stress shows a fair degree of anisotropy; see, for example, the data of ref. [72] for the PWA1480 single-crystal superalloy (Figure 3.72). Such experiments indicate that the yield stress along $\langle 001 \rangle$ is greater than that along $\langle 111 \rangle$; the increment $\Delta\sigma_y$ can be in excess of 200 MPa; under total strain control, the plastic strain along $\langle 001 \rangle$ is reduced by a factor $\Delta\sigma_y/E_{001}$ due to this effect, giving further justification to the superior properties along $\langle 001 \rangle$. Under load control, the situation is more complicated. The low modulus along $\langle 001 \rangle$ might at first be expected to result in large strains and poor fatigue life, but this is not observed in practice. Clearly, under load-controlled conditions one must be relying on the larger flow stress along that direction, or, alternatively, the life is limited by stress concentrations which behave in a strain-controlled manner.

Under LCF conditions, the cyclic stress–strain hysteresis loops show a number of interesting characteristics. First, the width of the loops is larger when high inelastic strains are imposed, which occurs for the stiffest orientations. See, for example, Figure 3.73 [71], for the case of $R = 0$. Second, a considerable degree of tension–compression yield stress

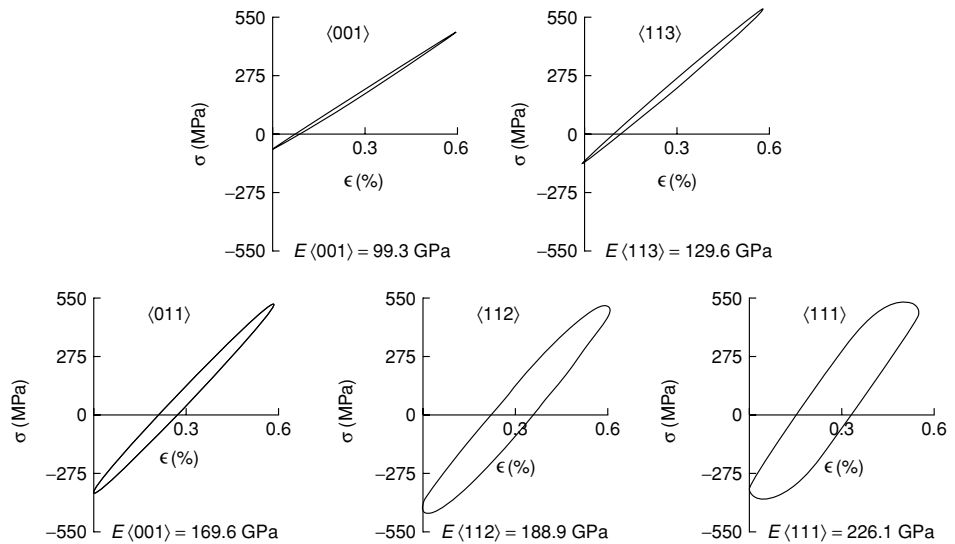


Fig. 3.73. Effect of crystallographic orientation on fatigue hysteresis loops for an experimental single-crystal superalloy at 980 °C and with $\Delta\epsilon_{\text{total}} = 0.6$, $R = 0$ [71].

asymmetry is observed for $\langle 001 \rangle$ orientations, with the yield stress in tension exceeding that in compression; Gabb *et al.* [73] report for Rene N4 values of 956 MPa and 818 MPa for tension and compression, respectively, at 760 °C. Interestingly, along $\langle 011 \rangle$ the yield stress is greater in compression than in tension – the values reported are 748 MPa and 905 MPa for tension and compression, respectively; along $\langle 111 \rangle$ little anisotropy was observed – 817 MPa and 842 MPa, respectively. Finally, the cyclic stress–strain curves show a considerable strain-softening response [74] – the values of σ_{max} and σ_{min} decline during the test consistent with a creep contribution inducing damage into the material. The experiments by Mughrabi *et al.* on the CMSX-6 superalloy (see Figure 3.74) indicate that the strain softening is most pronounced at higher temperatures, when the γ/γ' structure evolves via the rafting effect. These same workers have demonstrated that prestraining in tension, which induces rafts aligned normal to the stress axis, weakens the material in fatigue; conversely, prestraining in compression strengthens it.

As expected, the LCF life decreases quickly with increasing temperature. Figure 3.75 shows data [75] for the CMSX-4 single-crystal superalloy, measured under load control, with $R = 0$. One can see that no distinct endurance limit is displayed. Extensive shear banding and PSBs are found in the vicinity of the fracture surface; see Figure 3.76.

B High-cycle fatigue

The natural resonant vibratory frequencies of single-crystal turbine blading range from a few to several thousand hertz. Consequently, if the excitation frequencies due to turbulent flow around the aerofoil are in this range, then high-cycle fatigue (HCF) failure will occur. HCF loads are superimposed on the steady-state mean stresses arising from centrifugal

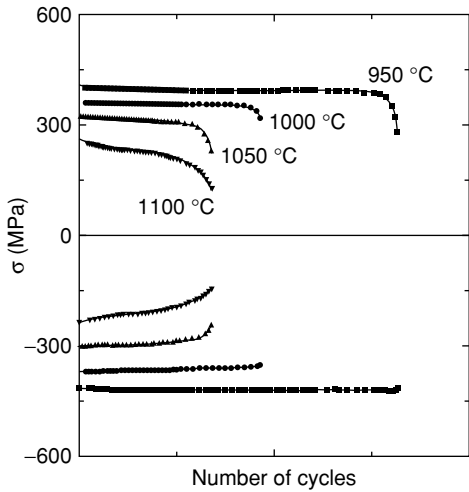


Fig. 3.74. Isothermal, cyclic fatigue deformation data for the CMSX-6 superalloy [74], showing the tensile and compressive peak stresses versus cycle number. Total strain range $\Delta\epsilon_{\text{total}} = 10^{-2}$; fully reversed conditions.

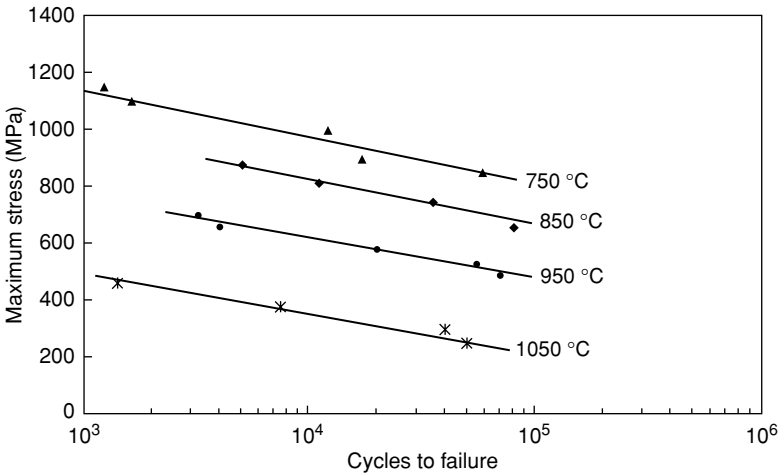


Fig. 3.75. Load control, LCF fatigue data [75] for the uncoated, second-generation superalloy CMSX-4 at various temperatures, with $R = 0$.

loading of the blade and thermal loads from thermal gradients set up by the action of the hot gas and the cooling air passing through the blading. Clearly, one then expects the HCF performance to depend on the R -ratio, and a considerable interaction of fatigue and creep damage mechanisms might be anticipated when the temperature is high and as $R \rightarrow 1$.

Recently, Wright and co-workers [76] have completed elegant and careful work which sheds light on the effects occurring during HCF loading. Load-controlled testing was carried out at 1038 °C on $\langle 001 \rangle$ -oriented single-crystal specimens of PWA1484 protected with a platinum aluminide coating. A first set of tests were carried out at 59 Hz for R -ratios of -1.0 , -0.33 , 0.1 , 0.5 , 0.8 and 1.0 ; obviously, when $R = 1.0$ there is no alternating component so that, in that limiting case, one is executing a constant load creep test. A second set of tests

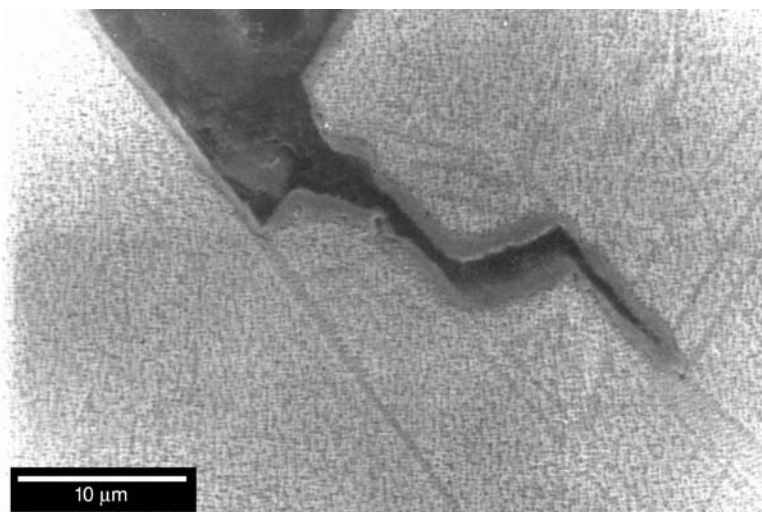


Fig. 3.76. Intense shear banding and persistent slip bands near the fracture surface of a CMSX-4 testpiece, cycled under constant load control at 850 °C, $R = 0$ and with $\sigma_{\max} = 760$ MPa [75].

were carried out at $R = -1.0$ and $R = 0.1$ at a range of frequencies from 59 Hz to 900 Hz. The results for the fatigue life, N_f , are plotted against the stress range, $\Delta\sigma$, on a conventional σ – N diagram (Figure 3.77(a)); for each R -ratio the Coffin–Manson relationship is obeyed. No evidence of an endurance limit was found. Moreover, the fatigue capability is seen to decrease, and the slope of the curve is steeper as $R \rightarrow 1$. In Figure 3.77(b), the data are plotted instead as σ_{mean} against time to failure. Here, the data for the higher R -ratios (0.5, 0.8 and 1.0) collapse along a single line defined by the creep rupture capability. This implies that life is independent of the alternating stress $\Delta\sigma$; thus the dominant mode of damage is creep. The data for lower R -ratios fall at lower times, implying that a fatigue mode is operative in that regime. The data at varying frequencies allow the fatigue and creep components to be further distinguished. When $R = -1$ the results were found to be independent of frequency for the range examined (see Figure 3.78(a)); this implies that a time-independent fatigue process is occurring at low mean stresses. At higher mean stresses, corresponding to $R = 0.1$, life is larger at the higher frequencies for any given $\Delta\sigma$; see Figure 3.78(b). This is because the failure depends more heavily upon the creep component which is time- rather than cycle-dependent. The characteristics of the fracture surfaces were consistent with this interpretation. At low mean stresses, corresponding to $R = -1$, failure was initiated either internally at pores, carbides or eutectic γ' , and crystallographic facets could be identified consistent with a fatigue mode of damage accumulation. At higher R -ratios, the appearance of the fracture surface resembled that of a creep rupture failure; it was rough with multiple surface cracking and a degree of necking was evident.

Wright and co-workers [76] displayed their results on the so-called Goodman diagram, on which the stress range, $\Delta\sigma$, is plotted against the mean stress, σ_{mean} , for a constant HCF capability. Figure 3.79 shows a typical diagram for 10^7 cycles to failure; the numbers correspond to the percentage creep damage determined from a creep–fatigue damage

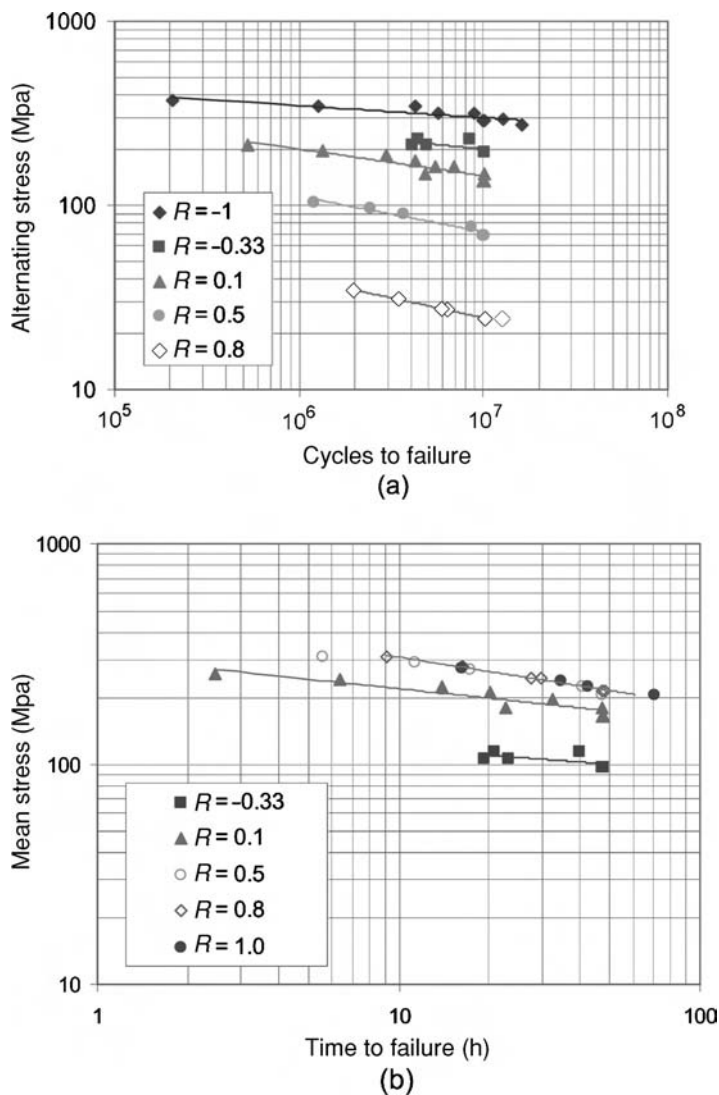


Fig. 3.77. HCF data for coated PW1484 at 1038 °C at 59 Hz, showing the effect of R -ratio: (a) conventional $\sigma-N$ curve, and (b) plot of mean stress versus time to failure [76].

mechanics model to which the data were fitted. For $R > 0$ one can see that creep mode becomes significant.

3.4 Turbine blading: design of its size and shape

The previous sections in this chapter have been concerned with the processing of turbine blade aerofoils by investment casting, with the metallurgical principles controlling the optimum superalloy compositions and with the mechanical behaviour observed during

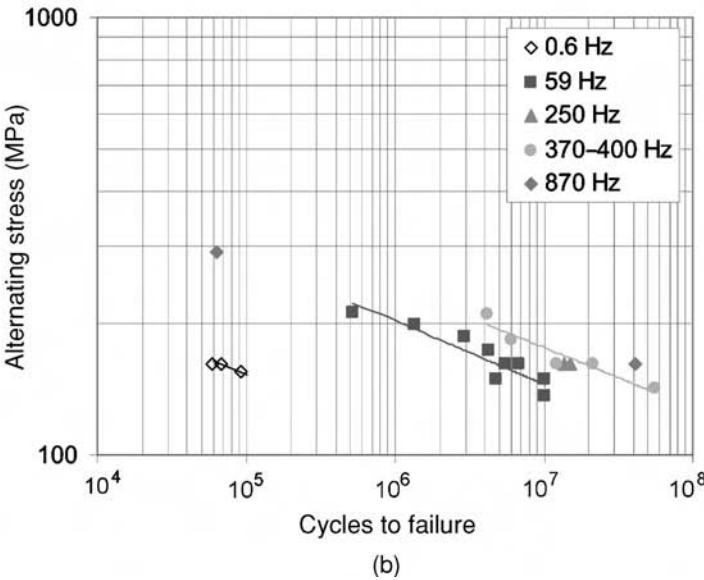
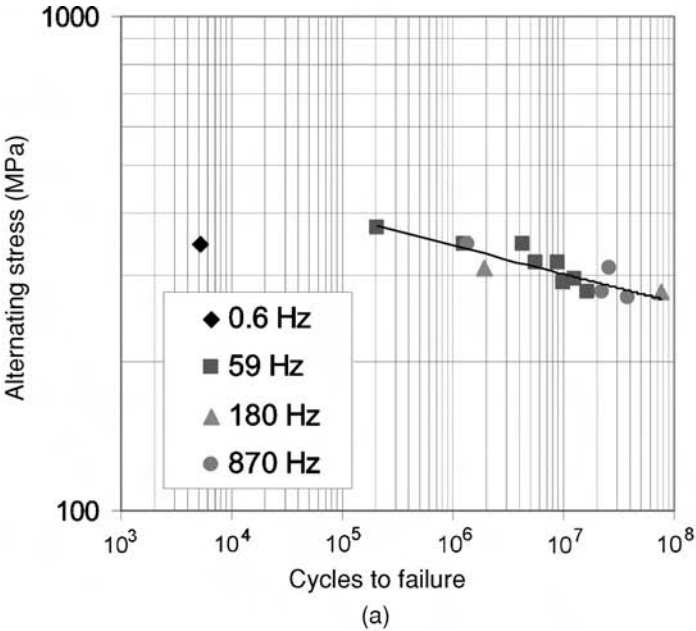


Fig. 3.78. HCF data for coated PW1484 at 1038 °C showing the effect of frequency: (a) $R = -1$ and (b) $R = 0.1$ [76].

service. Until now, the factors influencing the shape and geometry of the aerofoil have been largely ignored; these are dealt with in this section. It turns out that the relevant considerations depend largely on the laws of aerodynamics, fluid mechanics and thermodynamics [77,78].

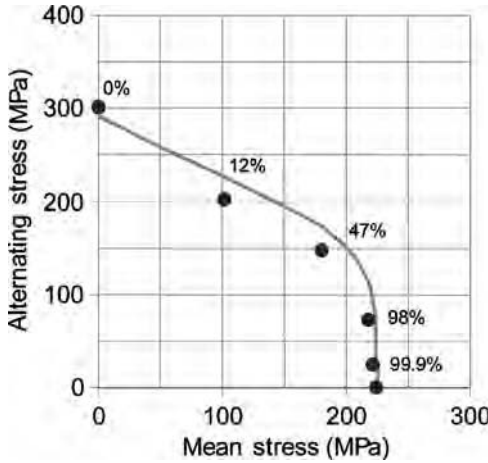


Fig. 3.79. Goodman diagram for coated PW1484 at 1038 °C at 59 Hz, 10^7 cycles [76]. The percentages are estimates of the percentage damage attributed to the creep damage mode.

3.4.1 Estimation of the length of the turbine aerofoils

As the hot gas expands through the turbine sections of the engine, its pressure decreases. Hence, to maintain the axial velocity at an acceptable value, it is necessary to *increase* the annulus or *throat* area, A , through which it flows past the nozzle guide vanes and blades; this is given by

$$A = 2\pi r_m l \quad (3.35)$$

where l is the length of the blading from root to tip and r_m is its mean radius measured in the radial direction from the axis of the engine. It follows that if r_m does not change substantially as the pressure drops across the various turbine stages, then l must increase; otherwise, an unacceptably high v_{jet} will be produced. Thus, necessarily the high-pressure (HP) turbine blades must be shorter than the low-pressure (LP) ones. An examination of a typical engine architecture (see Figure 1.1) confirms that this is indeed the case, although it is also clear that the r_m for the low-pressure stages is somewhat greater in practice than for the high-pressure stages.

For accurate calculations, the principles of fluid mechanics are required; only approximate estimates will be made here, since a detailed consideration is beyond the scope of this book. However, it turns out that the Mach number of the gas in a typical turbine is near 1.0, which is very much greater than the value of 0.3 below which it can be treated as incompressible; thus, rules for compressible flow must be invoked. Therefore, when considering the pressure and temperature of the gas stream, it is important to distinguish between the so-called static quantities, p and T , and the stagnation values, p_0 and T_0 (those which would be attained if the gas were brought to rest without work and heat transfer), via [77]

$$T_0/T = 1 + \left(\frac{\gamma - 1}{2} \right) M^2 \quad (3.36)$$

$$p_0/p = \left[1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right]^{\gamma/(\gamma-1)} \quad (3.37)$$

where M is the local Mach number, the ratio of the velocity, V , to the speed of sound given by $\sqrt{\gamma RT}$. As in Chapter 1, γ is the ratio of the specific heats at constant pressure and constant temperature, R is the gas constant and T is absolute temperature.

In order to determine the rate of mass flow, $\dot{m}$, through a section of area A , one must consider the combination

$$\dot{m}/A = \rho V \quad (3.38)$$

where ρ is the density of the gas stream and V is its velocity. Conservation of energy requires

$$\dot{m} c_p T + \dot{m} V^2/2 = \dot{m} c_p T_0 \quad (3.39)$$

where, upon rearranging, one has $V = \sqrt{2c_p(T_0 - T)}$. Assuming isentropic flow such that $\rho = CT^{1/(\gamma-1)}$, where C is a constant given by $C = \rho_0^{\gamma-1}/T_0^{\gamma-1}$, one has

$$\dot{m}/A = CT^{1/(1-\gamma)} \sqrt{2c_p(T_0 - T)} \quad (3.40)$$

After some algebraic manipulation, one can show that the dimensionless mass flow rate per unit area, Q , defined by the quantity $\dot{m} \sqrt{c_p T_0}/Ap_0$, is given by [77]

$$Q = \frac{\dot{m} \sqrt{c_p T_0}}{Ap_0} = \frac{\gamma}{\sqrt{\gamma-1}} M \left[1 + \left(\frac{\gamma-1}{2} \right) M^2 \right]^{-(\gamma+1)/(2(\gamma-1))} \quad (3.41)$$

One can see from the right-hand side that Q is constant for any given γ and M . Thus, this equation describes how the throat area, A , must change as p_0 and T_0 alter. The variation of Q with Mach number, M , is given in Figure 3.80, for the case $\gamma = 1.4$. It is found that the flow rate is a maximum at a Mach number of 1, when $Q = 1.281$; at this point, the cross-section, or nozzle, is said to be ‘choked’. For the turbine sections in a jet engine, this is usually a good assumption. It can be shown that, close to sonic velocity, very small changes in A produce large variations in the Mach number and pressure. Note that it has been assumed that the flow is one-dimensional and adiabatic. Provided that these are reasonable assumptions, Figure 3.80 can be used to determine A if estimates of the other quantities in the expression for Q are known. To a first approximation, one can take estimates of the pressure and temperature of the gas stream from the ideal thermodynamic cycle described in Chapter 1; this turns out to be not a bad assumption.

These considerations demonstrate that as p_0 decreases, A must increase – so the blades must get longer. The effect of the corresponding temperature drop is found to be rather less important in influencing A . This explains why the low-pressure blades of a gas turbine engine are very much longer than the high-pressure ones.

Example calculation

A high-pressure (HP) turbine stage passes a mass flow of 200 kg/s. If the stagnation pressure and temperature of the gas stream are taken as 1800 kPa and 1450 K, respectively, choose an appropriate annulus area, A . Assume that the flow is occurring close to Mach 1, so that the cross-section can be assumed to be choked. Hence, if the mean radius of the stage is to be 0.5 m, determine the length of the turbine blading. Take $c_p = 1000 \text{ J/(kg K)}$.

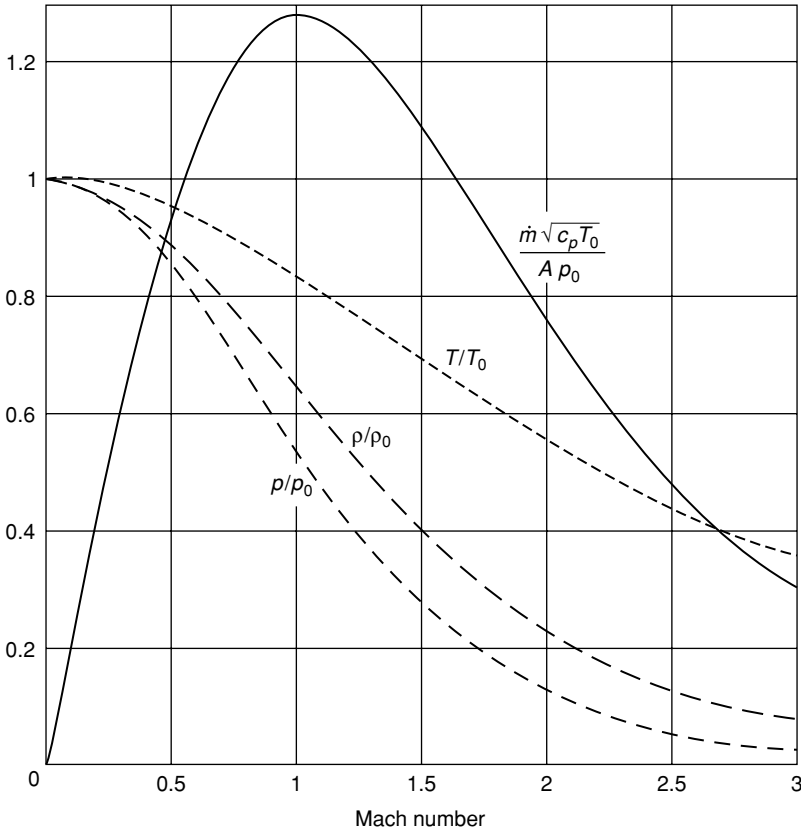


Fig. 3.80. Variation of the non-dimensional flow rate, $\dot{m} \sqrt{c_p T_0} / A p_0$, with Mach number, M , for a perfect gas with $\gamma = 1.4$, flowing through a uniform orifice. Also shown are the ratios of static to stagnation values of pressure, density and temperature. Adapted from ref. [77].

Answer At choking conditions, it follows that

$$Q = \frac{\dot{m} \sqrt{c_p T_0}}{A p_0} = 1.281$$

so that

$$A = \frac{200 \sqrt{1000 \times 1450}}{1.281 \times 1800 \times 10^3} = 0.1044 \text{ m}^2 = 2\pi r_m l$$

If $r_m = 0.5 \text{ m}$, it follows that $l = 33 \text{ mm}$.

3.4.2 Choice of mean radius for turbine blading

The function of the turbine blading is to extract mechanical work from the gas stream, and to do this as efficiently as possible. To a good approximation, the gas flow after exit from the nozzle guide vanes can be assumed to be axial; but, after interaction with the turbine blading, there is a significant tangential component.

Suppose that the flow enters at radius r (which remains unchanged) with tangential velocity $V_{\theta 1}$ and that it leaves with a value equal to $V_{\theta 2}$. The torque, T , required for this to occur is equal to the rate of change of angular momentum, and the power extracted, $\dot{W}$, is the torque times the angular velocity, Ω ; hence,

$$\dot{W} = T\Omega = \dot{m}\Omega r_m(V_{\theta 2} - V_{\theta 1}) = \dot{m}U_m(V_{\theta 2} - V_{\theta 1}) \quad (3.42)$$

where U_m is the speed at the mean blade radius and the product $U_m(V_{\theta 2} - V_{\theta 1})$ is known as the stagnation enthalpy; it is given the symbol Δh_0 .

Equation (3.42) is one form of the Euler equation. It can be written in non-dimensional form as

$$\frac{\Delta h_0}{U_m^2} = \frac{V_{\theta 2}}{U_m} - \frac{V_{\theta 1}}{U_m} = \frac{c_p \Delta T}{U_m^2} \quad (3.43)$$

where ΔT is the temperature drop across the stage. Extensive experimentation by aerodynamicists on turbine systems has shown that the non-dimensional work coefficient, $c_p \Delta T / U_m^2$, must not exceed ~ 2 , for satisfactory performance of the aerofoils.

Example calculation

A row of turbine blades is spinning at 6000 rev/min, and a temperature of 250 K is dropped across the stage. Estimate the mean radius, r_m , at which the row should be placed, for optimum efficiency. Take $c_p = 1000 \text{ J/(kg K)}$ as before.

Answer From the data given, $\Omega = 100 \text{ rev/s}$ and the core turbine work, $\Delta h_0 \sim c_p \Delta T$, is 250 kJ/kg. Now for an efficient turbine

$$\frac{c_p \Delta T}{U_m^2} \sim 2$$

hence $U_m = 350 \text{ m/s}$ and therefore $r_m = U_m / (2\pi \Omega)$ is approximately 0.56 m. It follows that if ω drops to half of its value, to 50 revs/s, r_m must be doubled to 1.1 m. Conversely, if the temperature drop is reduced by a factor of 2 to 125 K, then r_m is reduced by a factor of $\sqrt{2}$ to 0.40 m.

This result has some severe consequences for the design of the low-pressure (LP) turbine for a civil turbofan engine such as the Rolls-Royce Trent 800 or General Electric GE90. Since there is a need to restrict the tip speed of the fan blade to values not greatly exceeding the speed of sound, the LP shaft speed is relatively low – about 3000 rev/min – and therefore approximately one-third of the speed of the HP spool. Thus, given the considerations so far, r_m should ideally be made very large to return U_m and the work coefficient, $c_p \Delta T / U_m^2$, to acceptable values for optimum aerodynamic performance; however, calculations will confirm that this becomes difficult as the required r_m is then large enough to cause problems with the flow of the bypass air past the engine core.

In practice, this problem is solved by building a multistage LP turbine. Across each stage, a relatively small ΔT is dropped, thus restricting the work extracted. The penalty incurred, unfortunately, is greater weight – since the amount of turbomachinery required is then significantly increased – and greater manufacturing and maintenance costs. All large

civil turbofans suffer from this difficulty. For example, the three-spool Trent 800 has single-stage HP and intermediate pressure (IP) turbines, but a five-stage LP turbine. The two-spool GE90 is built with a two-stage HP turbine and a six-stage LP turbine. For Pratt & Whitney's two-spool PW4084, two HP stages are used, but a seven-stage LP turbine is required.

3.4.3 Estimation of exit angle from blade cross-section

The turbine consists of alternating rows of stationary guide vanes (stators) and blades (rotors); see Figure 3.81. The blades move in the tangential direction at a speed U_m , which, to a good approximation, is not very different from the local speed of sound. The function of the guide vanes is to remove the tangential component of the gas flow; the blades introduce this as a consequence of extracting work from the gas stream. Since it is not effective to allow the high-pressure gas exiting from the combustion chamber to undergo a single expansion across a single stage, the alternating rows of stationary vanes and moving blades facilitate a series of smaller expansions, which can be used efficiently [79].

In practice, the flow around the turbine blading is unsteady, and a full mathematical treatment is very complicated. However, it is helpful to consider the *relative* frame of reference local to the moving blades and the *absolute* reference frame pertinent to the stationary guide vanes. In the relative frame of reference, it has been found that the flow is steady enough to make calculations of sufficient accuracy for practical purposes. The situation is then as depicted in Figure 3.82. The gas is assumed to have an axial component of velocity given by V_x . The gas approaches the first row of stators with absolute velocity V_1^{abs} at an angle α_1^{abs} to the axial; its tangential component is thus $V_x \tan \alpha_1^{\text{abs}}$. After exit from the guide vane, the absolute velocity, V_2^{abs} , has increased, as has the tangential component, which is now given by $V_x \tan \alpha_2^{\text{abs}}$. Note that the absolute velocity of the gas exiting the guide vane is greater in the absolute frame than in the relative one; the converse is true for the gas exiting the blading – the angle α_3^{rel} is somewhat bigger than α_3^{abs} .

These considerations of so-called *velocity triangles* allow a first estimate of the shape of the trailing edge of the turbine blading, as the following example will demonstrate.

Example calculation

A first-row HP turbine stage is operating in the gas stream of a gas turbine engine. Estimate the flow direction which the gas should take out of the blading if the mean velocity, U_m , is set at twice the axial velocity of the gas stream.

Answer The stagnation enthalpy, Δh_0 , is given by

$$\Delta h_0 = U_m V_x (\tan\{\alpha_3^{\text{rel}}\} - \tan\{\alpha_2^{\text{rel}}\}) = U_m V_x (\tan\{\alpha_2^{\text{abs}}\} - \tan\{\alpha_3^{\text{abs}}\})$$

Assuming that the velocity of the gas relative to the blading is perfectly axial on exit from the nozzle guide vane, $\tan\{\alpha_2^{\text{rel}}\} = 0$. With $\Delta h_0/U_m^2 = 2$ and $U_m = 2V_x$ as given, then $\tan\{\alpha_3^{\text{rel}}\} = 4$ so that $\alpha_3^{\text{rel}} = 76^\circ$.

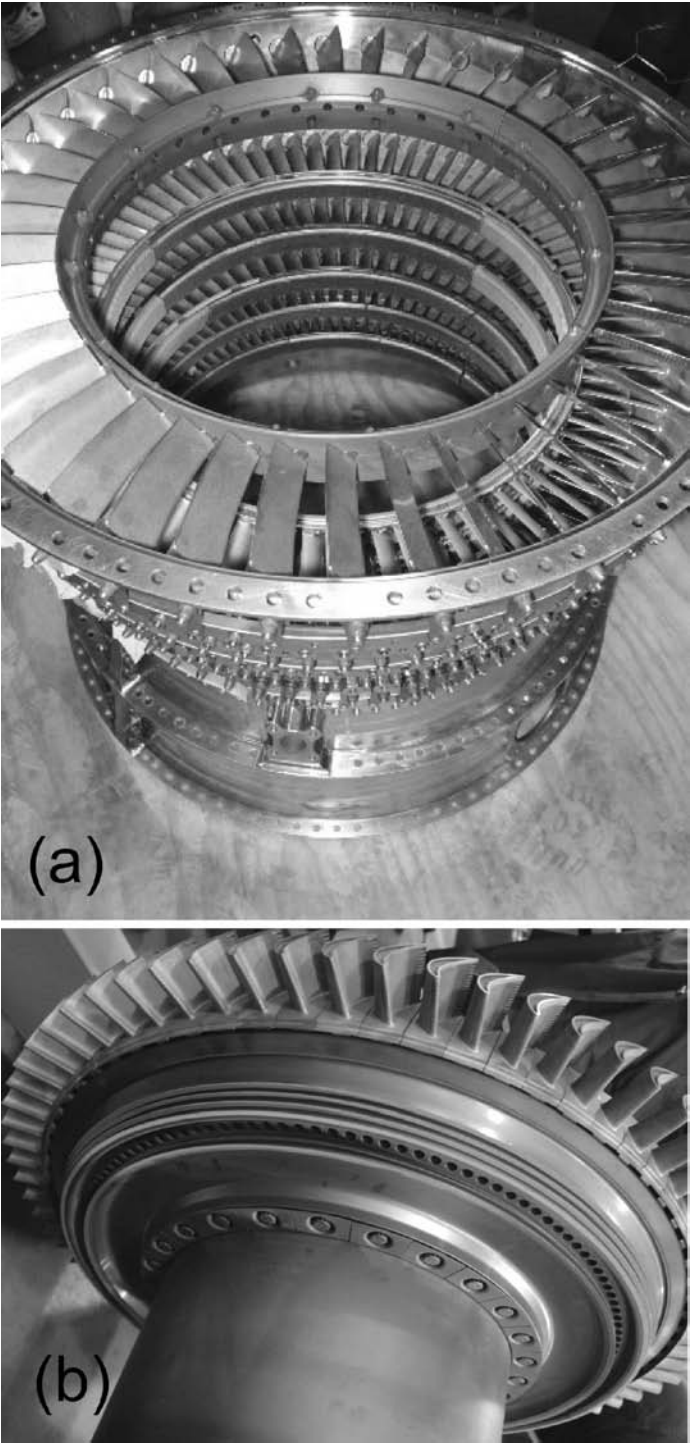


Fig. 3.81. Photographs of turbine components for the CFM-56 engine: (a) guide vanes (stators) in low-pressure turbine section, and (b) spool of high-pressure turbine blading.

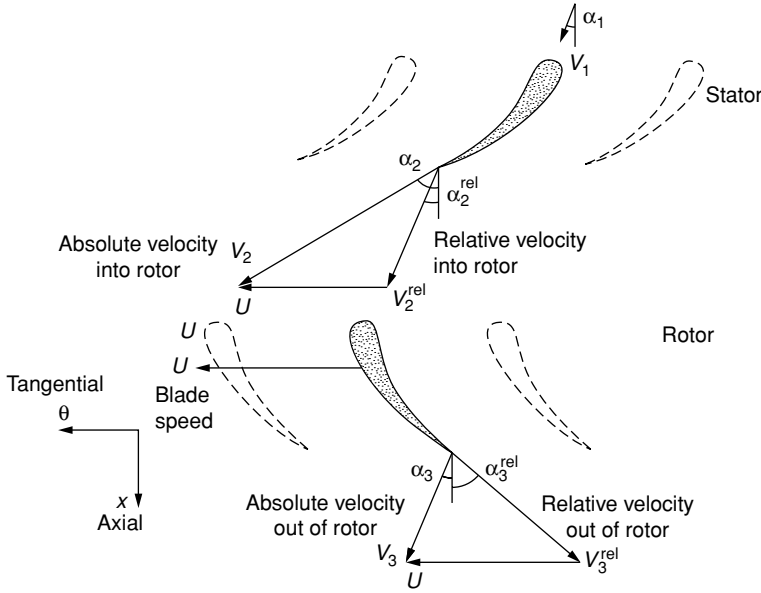


Fig. 3.82. Stage of an axial flow turbine, showing nozzle guide vanes (stators) and blading (rotors) in cross-section [77].

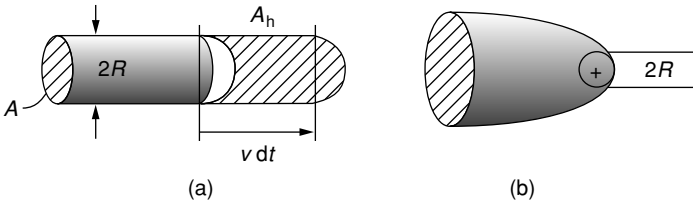


Fig. 3.83. Approximation of the dendrite tip as a cylinder with a hemispherical tip. (a) The cross-section of the cylinder, $A = \pi R^2$, determines the volume which grows in time dt . The surface area of the hemispherical cap, $A_h = 2\pi R^2$, determines the amount of radial solute diffusion. (b) The size of the dendrite in relation to the cylinder. After ref. [22].

Appendix. Growth of an isolated dendrite, using hemispherical needle approximation

The tip of a dendrite moving at a velocity v can be approximated as a cylinder with a hemispherical cap of radius R ; see Figure 3.83. This allows some important factors affecting growth to be accounted for.

The cross-section of the cylinder, $A = \pi R^2$, determines the volume which grows in time dt ; this is responsible for the rejection of solute into the melt. Likewise, the surface area of the cap, $A_h = 2\pi R^2$, determines the amount of radial solute diffusion. The flux, J_1 , due to solute rejection and the flux, J_2 , due to diffusion in the liquid ahead of the tip are given by

$$J_1 = Av(c_1^* - c_s^*) \quad J_2 = -DA_h \frac{dc}{dr} \Big|_{r=R} \quad (A1)$$

and, under steady-state conditions, these two fluxes must be equal.

The concentration gradient at the tip can be approximated by the value appropriate for a growing sphere, evolving under steady-state conditions:

$$\left. \frac{dc}{dr} \right|_{r=R} = -\frac{c_1^* - c_0}{R} \quad (\text{A2})$$

This can be proved by finding the solution to Laplace's equation in spherical coordinates:

$$\frac{d}{dr} \left(r^2 \frac{dc}{dr} \right) = 0 \quad (\text{A3})$$

by satisfying the boundary conditions and by evaluating the first derivative at $r = R$.

Combining the various expressions, one has

$$\frac{vR}{2D} = \frac{c_1^* - c_0}{c_1^* - c_s^*} \quad (\text{A4})$$

where the term on the right-hand side is known as the supersaturation, Ω . At constant Ω , the result predicts that the velocity, v , is inversely proportional to the dendrite tip radius, R .

Note that the expression for the flux at the dendrite tip is only approximate, since Laplace's equation was used. A rigorous analysis yields the Ivantsov solution, see ref. [22], but it can be shown that the approximation given is accurate in the limit $\Omega \sim 0$.

Questions

- 3.1 The materials used for high-pressure turbine blade aerofoils are often referred to as *single-crystal superalloys*. Explain why the use of the term 'single-crystal' is disingenuous.
- 3.2 How might the (a) axial and (b) transverse temperature gradients compare during the investment casting of a typical high-pressure blade for (i) an aeroengine and (ii) an industrial gas turbine?
- 3.3 During investment casting of hollow turbine blading, platinum pins are sometimes used to keep the internal ceramic and shell from moving relative to each other. These are expensive: why does one bother with them?
- 3.4 During solidification of a single-crystal alloy, the variation of fraction solid with temperature is an important consideration. Sketch the variation which might be expected assuming (i) equilibrium freezing conditions and (ii) Scheil freezing with no back-diffusion.

In order to assess whether back-diffusion is important, it has been suggested that one consider the parameter

$$\alpha = \frac{D_s t_f}{L^2}$$

where D_s is the diffusion coefficient in the solidifying phase, L is a characteristic distance equivalent to the dendrite arm spacing and t_f is the solidification time ($G/(v\Delta T)$), where ΔT is the solidification range. Using appropriate values of the G , v and ΔT for directional solidification, estimate the solidification time, t_f .

In order to determine whether back-diffusion is important, one should consider the Clyne–Kurz parameter, α' , where

$$\alpha' = \alpha \left(1 - \exp \left(-\frac{1}{\alpha} \right) \right) - 0.5 \exp \left(-\frac{1}{2\alpha} \right)$$

Note that $\alpha' = 0$ for Scheil solidification and $\alpha' = 0.5$ for equilibrium solidification given by the lever rule. Using this expression, and making estimates for the various parameters, determine whether back-diffusion is likely to be an important consideration for these alloys.

- 3.5 Investment cast components often exhibit dramatic changes in cross-section. How would one expect the spacing of isotherms to change when the cross-section of the casting increases in this way? Rationalise the observation that freckle chains are often found near re-entrant features, such as shrouds and platforms.
- 3.6 The preferred growth direction of a single-crystal superalloy is $\langle 100 \rangle$. Why?
- 3.7 Consider a typical Bridgman arrangement for the casting of directionally solidified components. What would be the effect of increasing the withdrawal velocity on the temperature gradient at the liquidus? What is the disadvantage of a more rapid withdrawal rate?
- 3.8 The following defects can occur during the casting of single-crystal components: (i) high-angle grain boundaries, (ii) freckles and (iii) spurious grains. What is meant by these terms? Give a brief explanation of the origin of each effect.
- 3.9 Various empirical relationships have been derived to describe the susceptibility of a single-crystal superalloy to freckling. Expressions which have been proposed for a freckling index, F , include

$$\frac{W + \text{Re}}{\text{Ta}} \quad \text{or} \quad \frac{W + 1.2\text{Re}}{\text{Ta} + 1.5\text{Hf} + 0.5\text{Mo} - 0.5\text{Ti}}$$

where the alloy compositions are input in wt%. Is there any physical basis underlying their use? On the basis of the known mechanism of freckling, is the value of F required to be small or large if defect-free castings are to be produced?

Using typical compositions for the superalloys (wt%), rank the susceptibility of the first-, second- and third-generation alloys to freckle formation. Are your findings consistent with the observation that the later generations are more freckle prone?

Many alloy developers are considering additions of Ru to single-crystal superalloys. How should you modify the expression for F to account for the presence of this element?

- 3.10 Find a good estimate of the thermal expansion coefficient of a typical single-crystal superalloy used for a turbine aerofoil. Use this to estimate the shrinkage strain expected when a casting cools to ambient temperature. What are the implications of your findings, particularly for the design of the ceramic mould?
- 3.11 In recent years, attempts have been made to improve the thermal gradients typical of an investment casting furnace. Developments such as liquid metal cooling (LMC) and gas cooled casting (GCC) have been made. It has proven possible to reduce the

primary dendrite arm spacing by about a factor of 2, from about 500 μm to 250 μm . How much shorter might then one expect the heat treatment time to be?

- 3.12 During directional solidification, heat is lost by (i) conduction to the chill through the solidified portion of the ingot, and (ii) radiation to the surroundings via the emissivity of the ceramic shell. Since these two contributions to the total heat loss, q_{total} , are additive, one has

$$q_{\text{total}} = (h_c + h_r)(T - T_0), \quad \text{where} \quad h_c = \frac{\kappa}{L}, \quad h_r = \frac{\sigma(\epsilon T^4 - \alpha T_0^4)}{(T - T_0)}$$

where h_r and h_c are the radiative and conductive heat transfer coefficients, respectively, and L is the length of the solidified portion of the casting. The constants ϵ and α can be taken to be 0.5. The Stefan–Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4)$.

Using reasonable values for κ , T and T_0 , find the length of the solidified ingot at which the contributions to the heat loss from radiation and conduction are equal. Hence show that, for a casting of any significant size, the majority of heat is lost by radiation rather than by conduction.

- 3.13 After casting, components fabricated from single-crystal superalloys undergo a complicated heat treatment designed to remove the microsegregation inherited from the casting process. What would be the implications of not doing this?
- 3.14 For the single-crystal superalloy CMSX-4, the density of the liquid phase has been measured to be 7.754 g/cm^3 at the melting point; it varies at a rate $-0.0009 \text{ g}/\text{cm}^3 \text{ K}$ with temperature. Using this information, compare the freckling susceptibility of the first-, second- and third-generation single-crystal superalloys Rene N4, CMSX-4 and CMSX-10. Do this by evaluating the differences in density between the liquid at the melting point and that at a (lower) temperature at which solidification is 10% completed. Use the Thermo-Calc software for making the equilibrium calculations. What assumptions have been invoked?

Hence show that there is a tendency for the later generations of single-crystal alloys to be prone to freckling.

- 3.15 Demonstrate that, for a binary phase diagram which possesses a liquidus slope of m , the undercooling, ΔT , experienced by an alloy of mean composition c_0 at a supersaturation Ω close to zero is given by

$$\Delta T \sim m c_0 \Omega (1 - k)$$

where k is the solid/liquid partitioning coefficient. (Hint: Note that $\Delta T = (c_1^* - c_0)m$, where $c_1^* = c_0/(1 - \Omega(1 - k))$. The factor c_1^* is the solute concentration in the liquid at the solid/liquid interface.)

- 3.16 The alloy development programmes have contributed greatly to the improved creep strengths of modern single-crystal superalloy materials. What have been the physical principles which underpin this work?
- 3.17 Harada has carried out an elegant experiment in which the creep lifetimes of the single-crystal superalloys TMS-75 and TMS-82+ were measured as a function of the fraction of the γ' phase present. The alloys were cast such that the compositions

of the γ and γ' phases were on a common tie-line, so that the phase compositions remain invariant. The results from this experiment are given in Figure 3.29.

What do the results from this experiment tell us about the creep deformation behaviour of superalloys? Why is the maximum creep resistance not imparted at a 50% fraction of γ' ?

- 3.18 When designing a new single-crystal superalloy composition, what are the properties which are desirable? Give examples of trade-offs between the various properties that are required.
- 3.19 In Figure 3.33, the evolution of the lattice misfit, δ , in the single-crystal superalloy SC16 is plotted during temperature decrements of 100 °C between 1200 °C and 800 °C. It is found that, after the temperature is stabilised, some further time is required before δ reaches its equilibrium value. What are the physical effects occurring during the periods whilst equilibrium is being attained?
- 3.20 It has been reported that the rate of oxide formation in Al_2O_3 -forming single-crystal superalloys is greatly increased with additions of Ti to the alloy chemistry. Explain why this effect occurs.
- 3.21 The Cannon-Muskegon company has made great strides in recent years in the reduction of sulphur levels in the casting stock used for single-crystal blading. Levels are now at the 5 ppm level or lower, whereas ten years ago they were at about 40 ppm. What advantages does this confer?
- 3.22 Hot corrosion can severely degrade a turbine blade aerofoil. Under what conditions can this occur? In practice, one can distinguish between type I and type II hot corrosion. Distinguish between these two modes of attack, noting the principle reactions which are responsible for them.
- 3.23 In designing the second-generation single-crystal superalloy PWA1484, Pratt & Whitney improved the oxidation resistance of the alloy by about a factor of 1.6 when compared against their first-generation alloy PWA1480. Compare the chemical compositions of these two alloys, and hence explain how this improvement was achieved.
- 3.24 Use Thermo-Calc to investigate the manner in which the elements in a typical nickel-based superalloy influence the size of the heat-treatment window. Develop an empirical equation such as

$$T_{\text{window}}(^{\circ}\text{C}) = A + B \text{ wt}\% \text{Al} + C \text{ wt}\% \text{Ta} + D \text{ wt}\% \text{W} + \dots$$

where A, B, C, . . . are parameters to be determined. Base the calculations on a typical second-generation superalloy. Once completed, correlate the sign and size of the parameters with the efficacy of the corresponding element to promote the γ' phase. Do any trends emerge?

- 3.25 Explain why it might not be sensible, even for single-crystal superalloys, to eliminate completely the grain-boundary strengtheners such as carbon and boron from the melt chemistry.
- 3.26 The gamma prime, γ' , phase in nickel-based superalloys is often cuboidal in form. Why?

- 3.27 At temperatures of around 950 °C, creep deformation of single-crystal superalloys occurs by thermally activated slip of octahedral dislocations on the {111}(110) system. However, the Schmid factor for this system is not a maximum on the (100) pole; rather, the maximum is at angle of $\sim 9^\circ$ from (100) on the (100)/(111) axis of the stereographic triangle. Quantify the percentage difference in applied stress level which can be tolerated in two single components: the first aligned exactly along (100) and the second at the point of maximum Schmid factor.
- 3.28 A single-crystal superalloy contains a volume fraction of 70% of the γ' phase. The misfit between the γ and γ' phases is 0.01. Estimate the inelastic (creep) strain necessary to relax the γ/γ' coherency stresses in this alloy.
- 3.29 Why is it fortuitous, particularly in terms of the mechanical properties of single-crystal blading, that the growth direction (100) is the least stiff from an elastic point of view?
- 3.30 A turbine blade cast from a polycrystalline nickel-based superalloy (density 8900 kg/m³) is working at a temperature of 600 °C. The root and tip radii, r_r and r_t , of the blade are 0.2 and 0.25 m, respectively, measured from the centre of the rotation. The blade rotates at 10 000 rev/min. Show that the stress in an element a distance R from the centre of rotation of the blade is given by

$$\sigma = \int_R^{r_t} \rho \omega^2 r \, dr$$

Use this result to find the variation of the stress along the blade. From this derive an equation for the variation in strain rate along the blade, and hence the rate of change of the blade with time.

You may assume that the blade is made from a superalloy which creeps in steady-state according to

$$\dot{\epsilon} = A\sigma \exp \left\{ -\frac{Q}{RT} \right\}$$

with $A = 4.3 \times 10^{-11} \text{ m}^2/(\text{N s})$ and $Q = 103 \text{ kJ/mol}$.

- 3.31 MacKay and Meier have measured the orientation dependence of primary creep performance of MarM247 single-crystals for various locations in the standard triangle of the stereographic projection. The (111) orientation performs best, followed by (001); (011) displays the worst behaviour. Rationalise this observation by determining the Schmid factors for deformation on the $(11\bar{2})\{111\}$ and $(1\bar{1}0)\{111\}$ slip systems. As a further exercise, plot out the contours of the Schmid factor within the standard triangle for the two slip systems.
- 3.32 The creep performance of three generations of single-crystal superalloys can be compared by plotting data for the stress, temperature and rupture life on a Larson–Miller plot; see Figure 1.16.

The turbine blading of a modern jet engine experiences temperatures of 1000 °C and stresses of 175 MPa during take-off. Compare the rupture lives of turbine blades cast in SRR99, CMSX-4 and RR3000. Take $C = 20$. Thus convince yourself of the

considerable advances in creep performance which have been made by the alloy designers.

At temperatures beyond 1050 °C the performance of RR3000 is only marginally superior to that of CMSX-4. What is the reason for this?

- 3.33 In this chapter, a model was presented for the evolution of the tertiary creep strain during deformation (see Equation (3.30)). Using the equations given, demonstrate that the creep strain is proportional to $\exp\{kt\}$, i.e. that it grows exponentially with time.

Unfortunately, one difficulty with the model is that the time to infinite strain is indeterminate, which means that it cannot be used to model data for time to rupture. Convince yourself that this is the case. Propose a modification to the analysis which circumvents this difficulty. In doing so, derive an expression for the time to rupture that is suitable for modelling purposes.

- 3.34 During the creep deformation of single-crystal superalloys at temperatures of around 750 °C, the $L1_2$ structure of γ' can be sheared by dislocations. Draw out a plan view of the $\{111\}$ plane for the $L1_2$ structure, and indicate on it the locations of the Ni and Al atoms. Suggest likely Burgers vectors for these dislocations in γ' and determine their magnitudes. Thus explain why $a/2\langle 1\bar{1}0\rangle\{111\}$ dislocations gliding in γ are unable to enter the γ' , unless the applied stress is very significant.
- 3.35 A typical single-crystal superalloy contains cuboidal γ' precipitates of edge length 0.5 μm . If the volume fraction of γ' particles is 70%, estimate the width of the γ channels which surround them. Hence convince yourself that any significant contribution to creep deformation made by $(111)\langle 1\bar{1}0\rangle$ dislocations in γ must involve significant cross-slip or climb, in addition to simple glide.
- 3.36 By modelling the microstructure of a single-crystal superalloy using springs of appropriate stiffnesses, compare the stress experienced by the horizontal and vertical channels of γ , under conditions of uniaxial loading. Use the dimensions determined in the previous question. Take E for γ to be 100 GPa and the value for γ' as 10% greater than this. Thus, rationalise the observation that creep deformation in these materials occurs first in the horizontal γ channels.
- 3.37 When single-crystal superalloys are tested at elevated temperatures under fatigue conditions, it is found that the fracture surface is either (a) quasi-brittle in form and characterised by crystallographic fracture along the $\{111\}$ planes, or else (b) ductile in form with the fracture surface perpendicular to the applied stress.

Giving your reasoning, which mode of failure is expected at (i) higher rather than lower temperatures, (ii) LCF rather than HCF conditions and (iii) lower mean stresses?

When the crystallographic mode dominates, the fatigue life is rather insensitive to the alloy content, i.e. the first-, second- and third-generation alloys do not behave very differently. However, the converse is true for the ductile mode. Account for this observation.

- 3.38 The strain-controlled LCF data of Dalal and co-workers are given in Figure 3.71, for an experimental single-crystal superalloy. Considerable fatigue anisotropy is displayed. Demonstrate that if one plots the product $\Delta\epsilon_{\text{total}}E_{\text{hkl}}$ rather than merely $\Delta\epsilon_{\text{total}}$ against

the N_f , where $\Delta\epsilon_{\text{total}}$ is the total strain range and N_f is the number of cycles to failure, then the fatigue data collapse onto a straight line. (Note that Dalal *et al.* report values of 97, 131, 165, 186 and 220 GPa for the Young's moduli, E_{hkl} , in the $\langle 001 \rangle$, $\langle 113 \rangle$, $\langle 011 \rangle$, $\langle 112 \rangle$ and $\langle 111 \rangle$ direction, respectively [71].)

- 3.39 The Goodman diagram accounts for the effect of mean stress, σ_{mean} , on the relationship between the fatigue life, N_f , and the alternating stress range, $\Delta\sigma$. An example is given in Figure 3.79 for the single-crystal superalloy PWA1484 under HCF conditions at 1038 °C for a lifetime of 10^7 cycles.

Is a greater temperature dependence expected for the creep rupture or the fatigue mode of failure in this case? Hence draw a sketch illustrating how the curves on the Goodman diagram are expected to alter as the temperature is changed. Will the gradient be greater or less as the temperature is increased? How will the behaviour of (i) coated and (ii) uncoated specimens compare?

- 3.40 Consider what happens to the stress distribution in a hollow turbine blade during engine start-up. The blade starts cold and heats up from outside in; therefore creep is expected first near the outer surfaces. Examine what effect this will have on the load borne by the internal (cooler) webs of the blading as creep proceeds.
- 3.41 The number of turbine blades required is dictated by the pressure drop to be taken across any given stage; in fact, these two numbers are inversely proportional to each other. A suitable equation based upon idealised work is as follows:

$$\frac{\dot{M}\Delta V_w}{N_b} = C_L(P_{\text{in}} - P_{\text{out}})hb$$

where $\dot{M}$ is the total mass flow of air in kg/s, ΔV_w is the velocity change induced by the blading, N_b is the number of blades of height h and breadth b , C_L is a lift coefficient of about 0.8 and P_{in} and P_{out} are the pressures of gas entering and leaving the turbine stage, respectively.

Using this information, estimate the number of turbine blades, N_b , required in the first stage of the Trent 800 high-pressure (HP) turbine, using typical dimensions for the blading and making assumptions where necessary.

What is the weight of a typical HP blade, denoted M_b ? Hence estimate the weight of a complete row of high-pressure turbine blades.

The following information is provided. The velocity of gas exiting from the nozzle guide vanes (NGVs) is about 0.3 km/s, and this can be assumed to be axial; the turbine blade accelerates it by a factor of $\sqrt{2}$. The pressure drop across the stage is about 800 kPa.

- 3.42 The HP turbine blades of Rolls-Royce's Trent 800 engine are shrouded, but in General Electric's GE90 they are shroudless. Estimate the weight penalty, and hence the extra loading on the bore of the turbine disc from the presence of the shrouds. Why might it not be sensible to remove them? List the advantages and disadvantages of shrouded and shroudless blades.
- 3.43 When designing the turbine stages of an aeroengine, weight must be minimised. However, to maintain efficiency, the non-dimensional loading of the turbine, given by $c_p\Delta T/U_m^2$, must not exceed 2; here ΔT is the temperature drop across the turbine

stage and U_m is the velocity of the blade at mean height. The heat capacity, c_p , has a value of 1000 J/(kg K) .

Use these considerations to find a suitable mean location for the HP and LP turbine blades, relative to the axis of the engine, given typical HP and LP shaft speeds and estimated values of ΔT . Establish whether there are benefits to designing a turbine consisting of multiple (two or more) stages, across which a corresponding fraction of ΔT is dropped.

- 3.44 In practice, a turbine blade might be considered to have ‘failed’ for a number of different reasons. What might these reasons be?
- 3.45 For the shroudless high-pressure blades commonly used in aeroengines manufactured in the USA, wear of the turbine tip and turbine seals can be considerable. What are the effects of this wear likely to be?
- 3.46 Convince yourself that, if a turbine blade is modelled as a uniform slab, the centrifugal stress at the blade root is given by $\rho\omega^2 h\bar{r}$, where $\bar{r}$ is the average of the blade root and tip radii, and that the peak bending stress can be estimated from

$$\frac{\sigma_{\text{peak, bending}}}{d/2} = \frac{\dot{M}\Delta V_w}{N_b} \left(\frac{2h}{3} \right) / bd^3/12$$

where b is the breadth, d the depth and h the height of the blade. The other symbols have the same meaning as in Question 3.41.

Suppose that after a preliminary turbine design is completed, the bending stresses are found to be too high. What simple measure could be taken to reduce them without significant alteration of the dimensions and weight of the blade?

- 3.47 The high-pressure (HP) stage of the CF6 turbine engine contains 82 blades. Each weighs about 200 g, although, in practice, each is not perfectly uniform – thus, each set of blades must be arranged on the turbine disc such that the turbomachinery is as balanced as possible. The mean radius, r_m , of the turbine blades is 0.5 m, measured from the axis of the turbine disc.

The degree of unbalance is expressed in units of g cm; thus 1 g of excess weight displaced 2 cm from this axis is equivalent to 2 g cm of unbalance. Write a computer program which generates a distribution of masses for the 82 blades, consistent with a standard deviation in the mass equivalent to 0.1%. Then determine the order in which the blades should be placed on the disc, if the unbalance is to be minimised. What magnitude of unbalance is expected?

- 3.48 Many of the latest generations of industrial gas turbines (for example, GE Power Systems’ ‘H’ class unit) employ combined-cycle operation such that closed-loop steam cooling is used to moderate the temperature of the high-pressure turbine blading. Explain why this situation provides an extra incentive for the use of single-crystal technology (for example, Rene N5) for the blading.
- 3.49 The Pratt & Whitney company uses the concept of a ‘technical readiness level’ (TRL) to define the status of a new material which is being considered for engine application. Thus, TR-1 corresponds to the ‘eureka’ moment, with TR-9 being the level just prior to service entry. During the development cycle of a new single-crystal turbine blade alloy, to what might the technical readiness levels TR-2 through to TR-8

correspond? Describe the work which needs to be done and the technical progress to be demonstrated during these periods.

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4 Superalloys for turbine disc applications

The turbine discs are amongst the most critical of components in the aeroengine. To put this into perspective, consider a typical modern civil turbofan such as the Trent 800 – the turbine discs represent about 20% of its total weight and their cost accounts for about 10% of the engine's value upon entry into service. For a military engine, such as the EJ200, the figures are nearer 5% and 25%. The primary function of the turbine discs is to provide fixturing for the turbine blades located in the gas stream, from which mechanical energy is extracted; the complete assembly of discs and blades is then capable of transmitting power to the fan and compressor sections, via the shafts which run along almost the complete length of the engine. During the design stages, their geometry must be optimised by balancing the competing demands of minimisation of weight, dimensional stability and mechanical integrity. Thus risk mitigation by an appropriate disc lifing strategy is vital, since a disc failure represents a potentially fatal hazard to the aircraft and its occupants. Metallurgical damage due to fatigue during service and its acceleration due to oxidation and/or corrosion must therefore be quantified and predicted, so that each disc is withdrawn from service after a prescribed number of take-off/landing cycles, known as the *safe working life*.

To convince oneself of the properties required of a turbine disc alloy, it is instructive to consider the operating conditions experienced by a typical high-pressure (HP) turbine disc. The HP turbine blades project directly into the gas stream in excess of 1550 °C, causing the temperatures in the rim of the turbine disc to approach 650 °C or beyond during service. The rotational speeds are very significant, the exact value tending to increase as the disc diameter decreases; for the Trent 800 engine 10 500 revolutions per minute is typical, whilst for helicopter engines the figure may be in excess of four times this. Consequently, the mechanical stresses generated in the bore region may reach 1000 MPa during take-off, i.e. they may exceed the uniaxial tensile yield strength of the material. Higher stresses will be present in the overspeed condition which arises if the shaft fails – a theoretical possibility which must be acknowledged in the design process. The energy stored in a HP turbine disc under such conditions is very significant: equivalent, in fact, to 0.75 kg of high explosive, or a saloon car travelling at over 100 mph.

4.1 Processing of the turbine disc alloys

The discs for the gas turbine engine are fabricated by the machining of superalloy forgings, but two distinct approaches are available for the preparation of the billets to be operated on

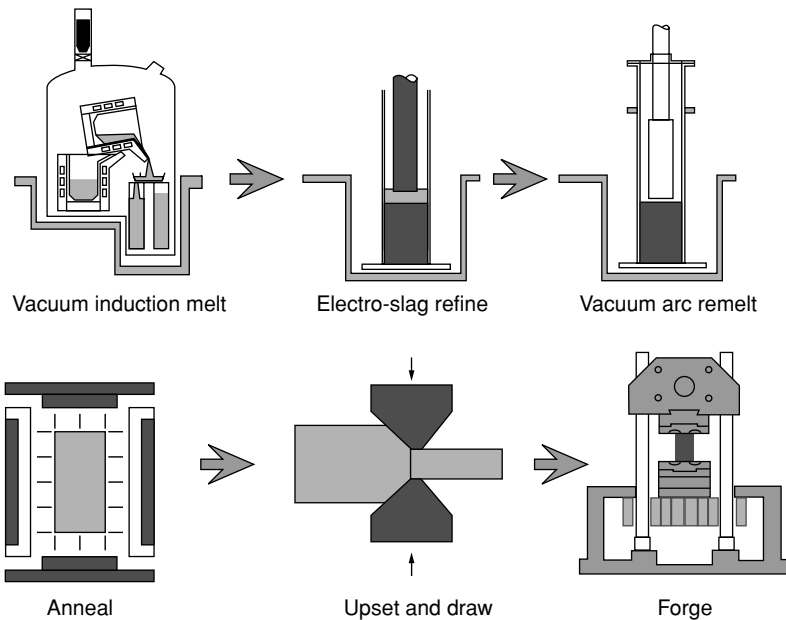


Fig. 4.1. The sequence of processes used for the production of turbine disc alloys by ingot metallurgy. Adapted from ref. [3].

by the forgemaster. The first is conventional ingot metallurgy, which involves the thermal-mechanical working of material produced by vacuum induction melting, electro-slag remelting and vacuum arc remelting [1]. Billets processed in this way are referred to as *cast-and-wrought* product. A second route involves powder metallurgy. The choice of route depends upon a number of factors, but is largely dictated by the chemistry of the chosen superalloy [2]. Alloys such as Waspaloy and IN718 are generally prepared by ingot metallurgy, since the levels of strengthening elements Al, Ti and Nb are relatively low, so that the additional cost associated with powder processing cannot be justified. Figure 4.1 illustrates the various steps taken in this case [3]. Vacuum induction melting is followed by electro-slag refining and vacuum arc remelting; after annealing to improve the compositional homogeneity, the billet is thermal-mechanically worked before undergoing a series of forging operations. Unfortunately, ingot metallurgy cannot be applied to the heavily alloyed grades such as Rene 95 and RR1000, since the levels of segregation arising during melt processing and the significant flow stress at temperature cause cracking during thermal-mechanical working; instead, powder-processing is preferred, see Figure 4.2 [3]. Here, vacuum induction melting (VIM) is used as before, followed by remelting and inert gas atomisation to produce powder – this is then sieved to remove any large non-metallic inclusions inherited from the processing. This step is important since it improves, in principle, the cleanliness of the product. To prepare a billet ready for forging, the powder is consolidated by sealing it into a can, which is then degassed and sealed, and hot isostatic pressing and/or extrusion follows. In principle therefore the concentration of inclusions will be lower in the powder metallurgy

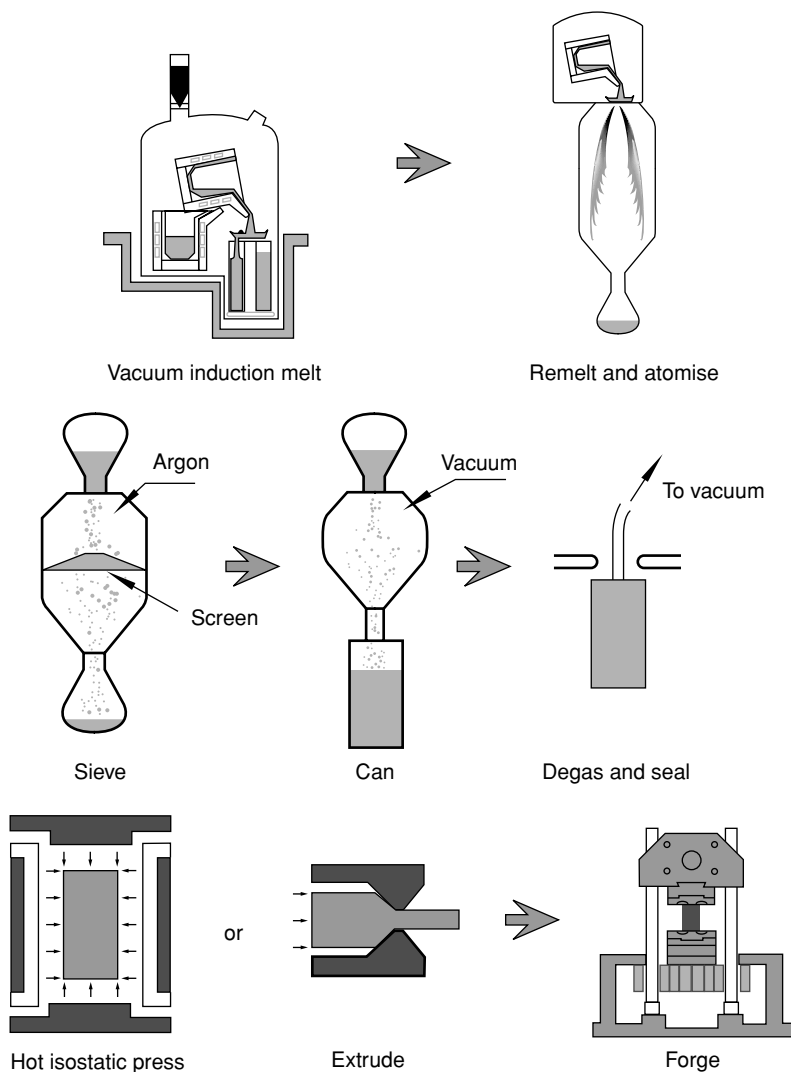


Fig. 4.2. The sequence of processes used for the production of turbine disc alloys by powder metallurgy. Adapted from ref. [3].

product than in that produced by ingot metallurgy, although this is achieved at greater cost and by a processing route which is significantly more complex.

Regardless of the route to be employed for the processing of the turbine disc material, VIM is used as the standard melting practice for the preparation of the superalloy stock. Vacuum metallurgy dates from the 1840s, but considerable advances were made in Germany during World War I when Ni–Cr alloys were produced as alternatives to platinum-based thermocouple alloys. The first commercial application of the superalloys came in the 1950s with the production of 5 kg heats of Waspaloy; each heat was forged to a single turbine blade. The success of the industry can be judged by noting that worldwide production has expanded

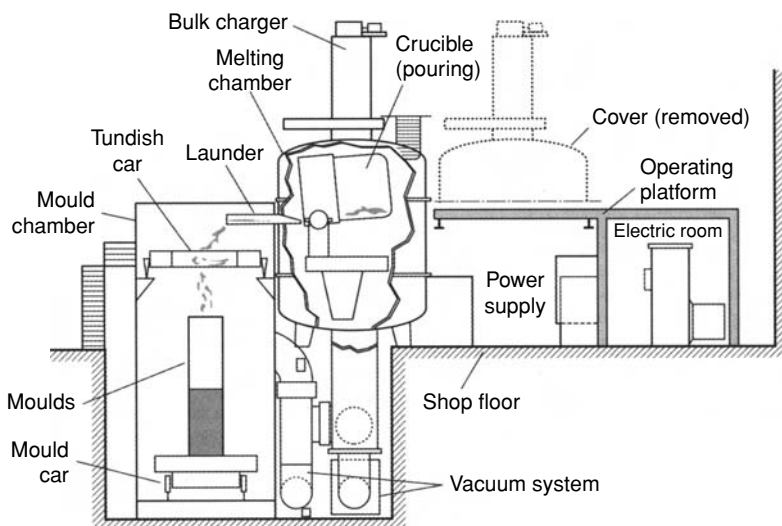


Fig. 4.3. Schematic illustration of the apparatus used for the vacuum induction melting of the superalloys. (Courtesy of Gern Maurer.)

from about 4 tonnes per annum in 1923 (largely in Germany) to greater than 30 000 tonnes in the year 2000. The general principles are as follows. Raw metallic materials including scrap are charged into a refractory crucible, which is surrounded by water-cooled copper coils which carry an electric current at a frequency of 50 or 60 Hz; the current must be chosen carefully since there is a need for sufficient heating and electromagnetic stirring, without excessive erosion of the refractory lining of the furnace. The ceramics ZrO_2 , MgO or Al_2O_3 are commonly employed as refractory crucibles, often in combination with one another; the composition 90% MgO /10% Al_2O_3 is common. Figure 4.3 illustrates a typical arrangement. The crucible is maintained under a vacuum of better than 10^{-4} atm during melting of the charge, allowing a series of chemistry adjustments to be made. Typically, more than 30 elements are refined or removed from the superalloy melt during VIM processing [3]. A final step involves the decanting of the liquid metal from the crucible into a pouring system, and the casting of it into moulds under a partial pressure of argon. Consequently, VIM is a batch, rather than a continuous, process.

A number of important phenomena occur during VIM processing. First, substantial reductions in the concentrations of nitrogen and oxygen are achieved, since these degas and are thus removed by the vacuum system. This helps to prevent the oxidation of reactive elements such as Al, Ti, Zr and Hf; this is probably the primary justification for vacuum processing. Second, low-melting-point contaminants such as Pb, Bi are removed from the melt by volatilisation [4] on account of their substantial vapour pressures; see Figure 4.4. These elements commonly enter the process via recycled material, for example, because they are used to mount superalloys during machining processes. The rate-controlling step here appears to be mass transport in the liquid phase – consequently, electromagnetic stirring accelerates the removal of these ‘tramp’ elements. This is fortuitous, since the high-temperature properties of the superalloys are substantially impaired if these elements

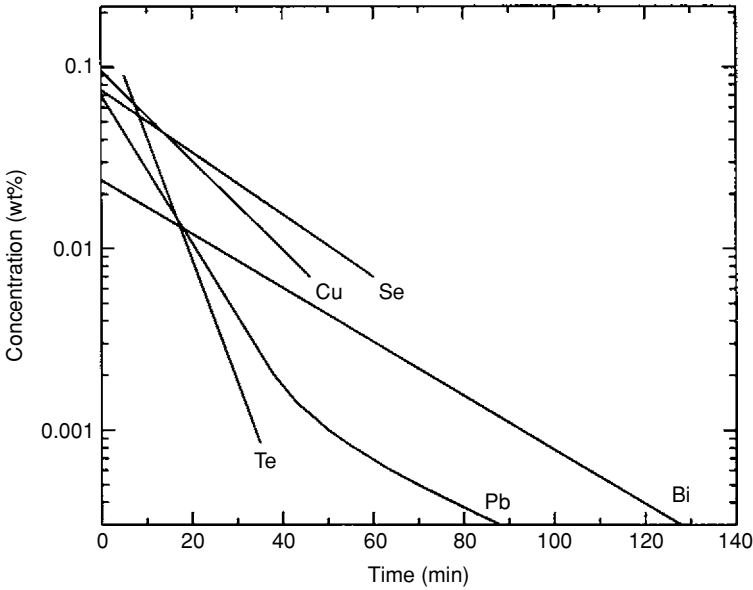


Fig. 4.4. Evaporation of elements from Ni-20 wt%Cr alloy in a vacuum induction melting furnace [4].

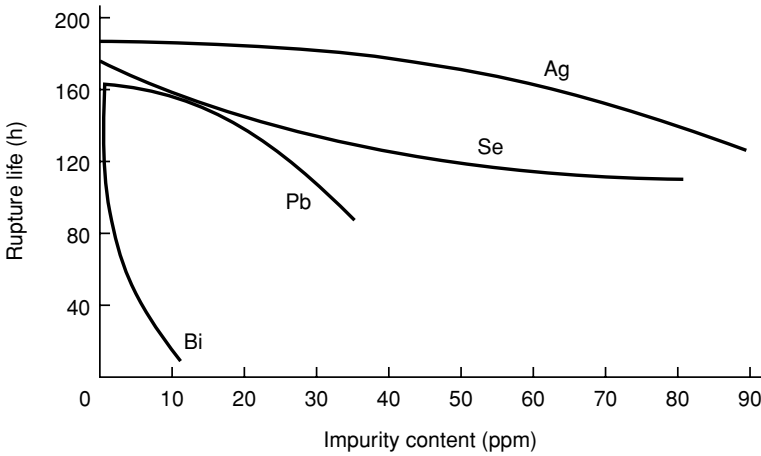


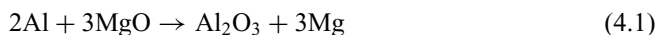
Fig. 4.5. Effect of various trace elements on the stress-rupture life of IN718 at 650 °C and 690 MPa. Data taken from ref. [5].

are present (see Figure 4.5); for IN718 tested in creep at 650 °C and 690 MPa, the rupture life is decreased ten-fold and two-fold by the addition of only 10 ppm of Bi and 40 ppm of Pb, respectively [5]. This explains why strict limits are placed on the levels of trace elements which are permissible in premium-grade superalloys, and justifies the emphasis placed on compositional analyses during processing for quality assurance purposes. Finally, desulphurisation of the melt is achieved typically by adding lime (CaO) and fluxing agents.

Removal of slag, which also comes about via deterioration of the crucible lining and the oxidised surfaces of revert material, is achieved mainly by flotation in launders built into the pouring system, but also using ceramic ZrO_2 -based filters; these prevent it from being incorporated into the solidified ingot.

4.1.1 Processing by the cast-and-wrought route

In practice, the VIM-cast material is not suitable for immediate use in gas turbine engines. First, the level of segregation is so extreme that thermal-mechanical working of the ingot would prove impossible should this be attempted. Second, the VIM ingot is not mechanically sound, since it contains a solidification pipe, cracks and porosity, which would need to be machined away causing unnecessary wastage. Finally, a population of refractory particles of ceramic is present; this needs to be reduced to more reasonable levels to ensure adequate fatigue resistance; see Section 4.3. These are introduced by erosion of the refractory lining and by reactions such as



where the reactants Al and MgO arise from the melt and lining, respectively. For these reasons, superalloy stock processed by conventional cast-and-wrought processing is subjected to further *secondary* melting processes such as vacuum arc remelting (VAR) and possibly also electro-slag remelting (ESR). These reduce ingot pipe and improve the yield and forgeability of the large ingots which are used for disc applications. For components which demand the highest possible levels of cleanliness, *triple*-melted product is available: this will have undergone VIM followed by ESR and finally VAR. Where triple melting is employed, VAR will follow ESR because the different heat transfer characteristics allow the production of a larger diameter ingot without causing melt-related defects such as freckles.

A Vacuum arc remelting

This involves the melting under vacuum of a consumable electrode into a copper-cooled crucible; see Figure 4.6. The electrode is a casting which will have been produced either by the VIM process or else the ESR process, which is described below. During processing, the energy necessary for melting is provided by a DC arc, which is struck between the electrode (the cathode) and the molten pool at the top of the solidifying ingot (the anode). The current used is typically $\sim 10\,000$ A, and the applied voltage is in the range 20 to 50 V. The scale of the process is impressive; finished VAR ingots are several metres in length and weigh up to 20 tonnes.

The heat and mass transfer processes occurring during melting have a profound influence on the quality of the resulting VAR ingot. During processing, a thin molten film is formed on the electrode, which flows under gravity to form hanging protuberances. Metal transfer occurs as droplets form and momentarily short circuit the arc as they fall from the molten film. To maintain a constant arc gap, a ram arrangement is used to drive the electrode into the VAR crucible as melting proceeds. Considerable emphasis is placed on control strategies and process monitoring. For example, dynamic load cell weighing systems have